

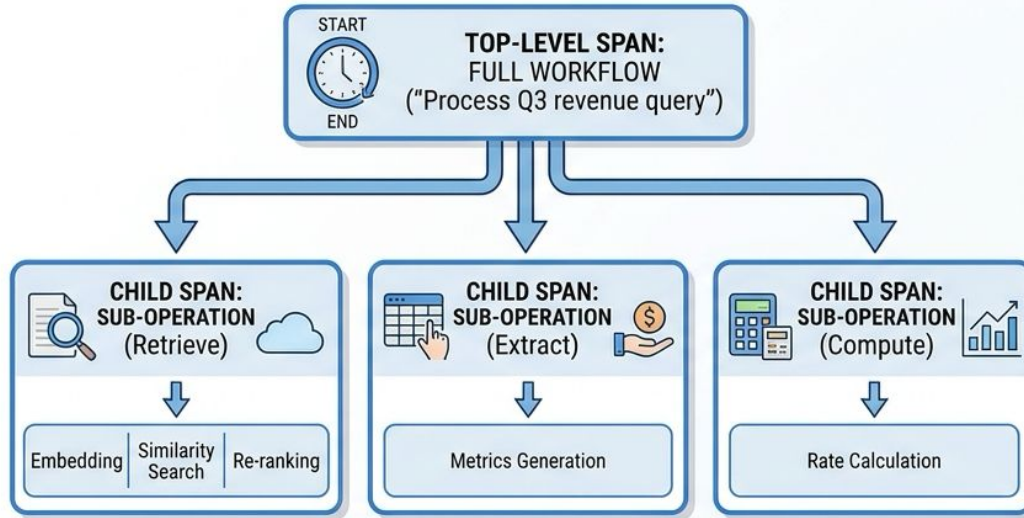


U.S. General Services
Administration

Chapter 3.6: Trace Analysis and Execution Debugging

US General Service Administration
AI Community of Practice - March 2026

Trace Architecture -- Hierarchical Spans



SPAN UNIT COMPONENTS

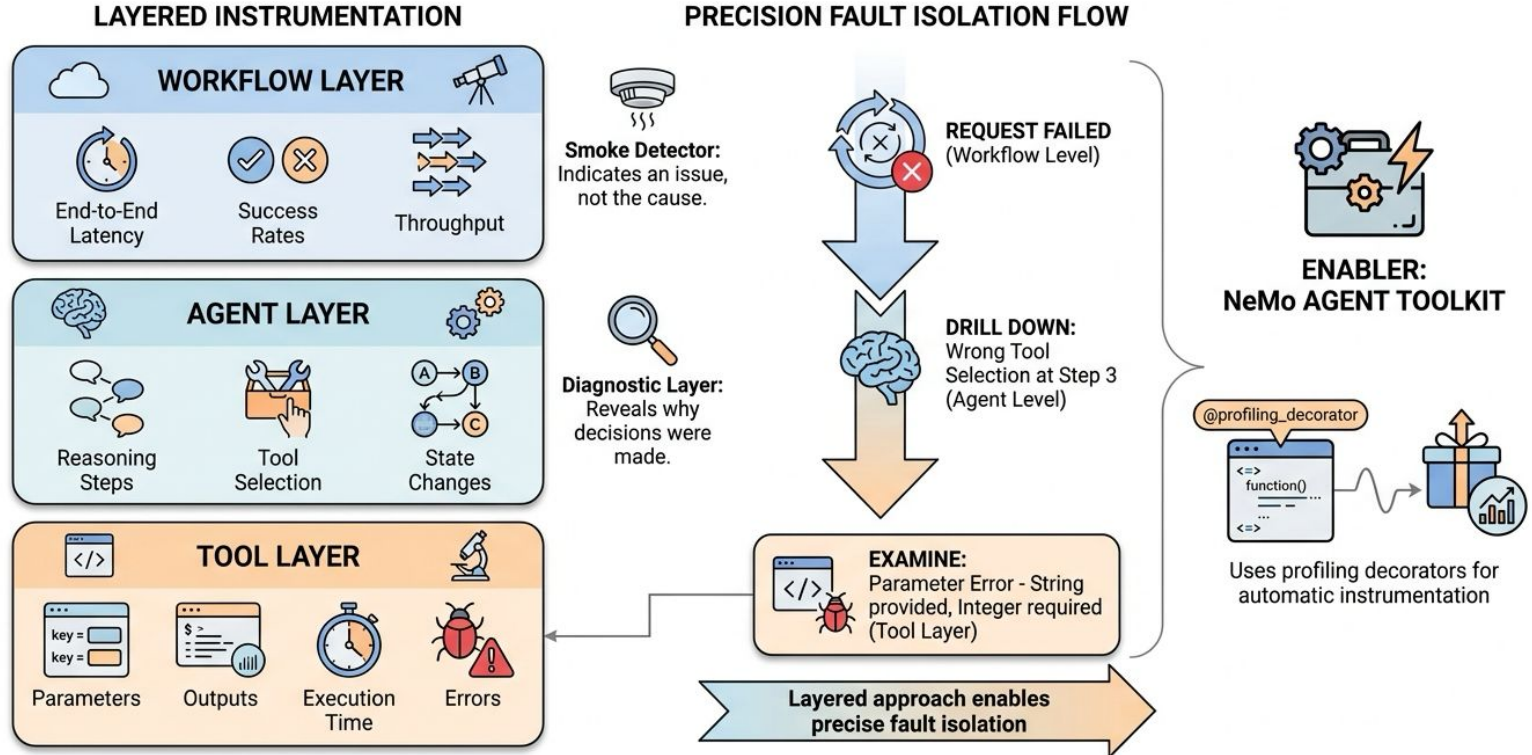


CAPTURES REASONING & THOUGHT PROCESS

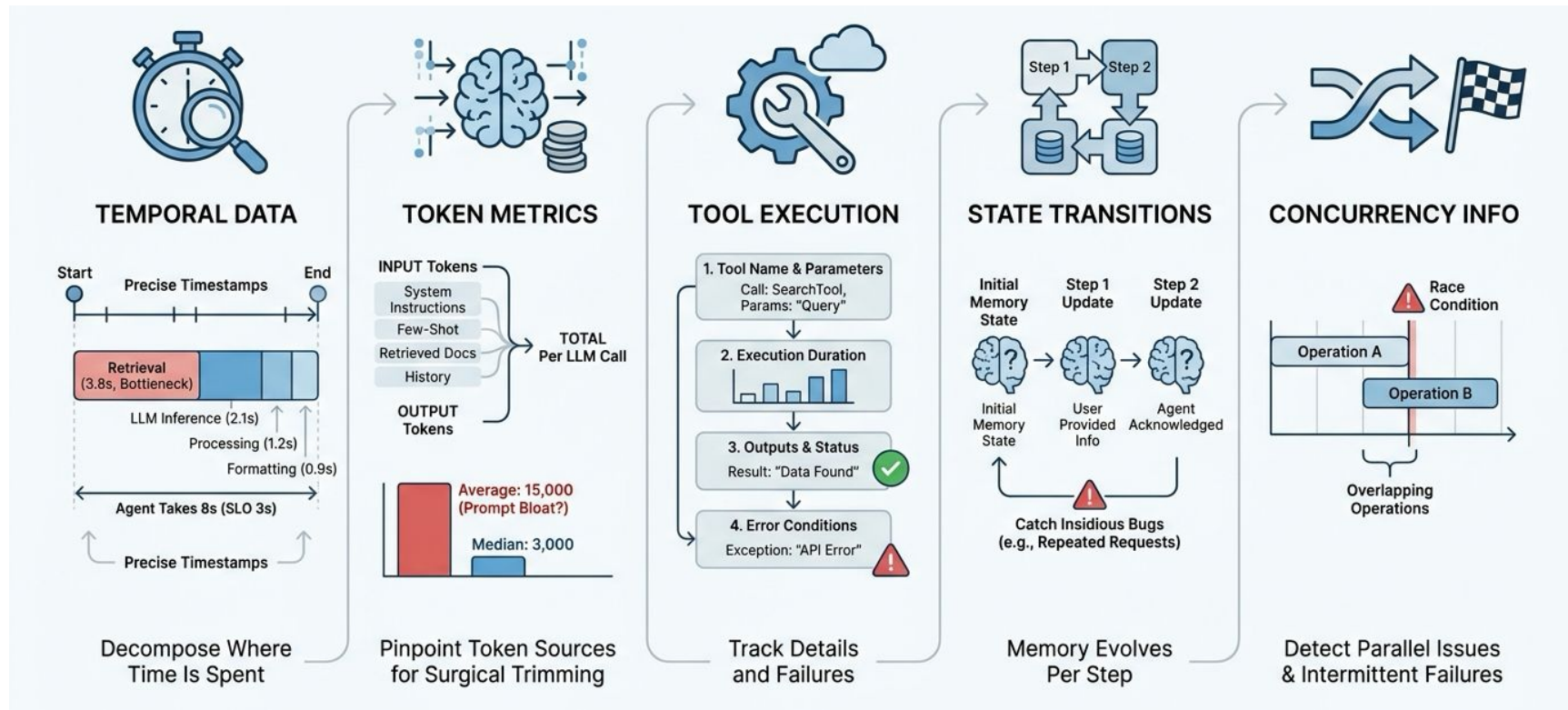


RECURSIVE NESTING:
ARBITRARY DEPTH

Three Layers of Instrumentation

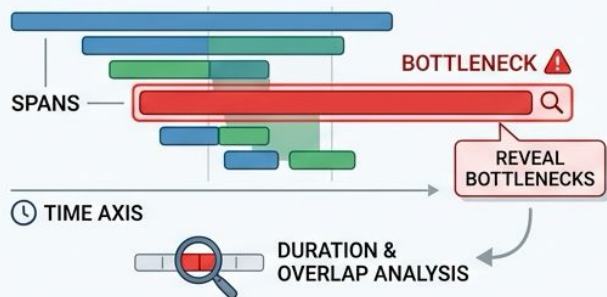


What Traces Capture -- The Essential Data Points

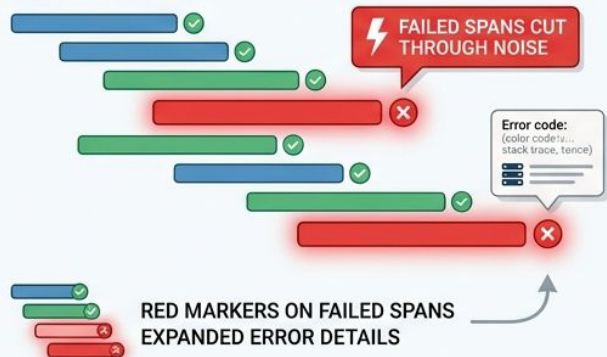


Trace Visualization -- Seeing the Invisible

TIMELINE VIEWS



ERROR HIGHLIGHTING



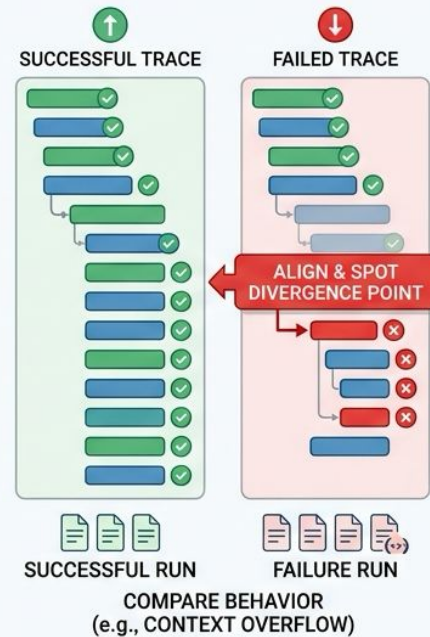
HIERARCHICAL DRILL-DOWN



FILTERING

BY TOOL BY LATENCY THRESHOLD BY ERROR TYPE

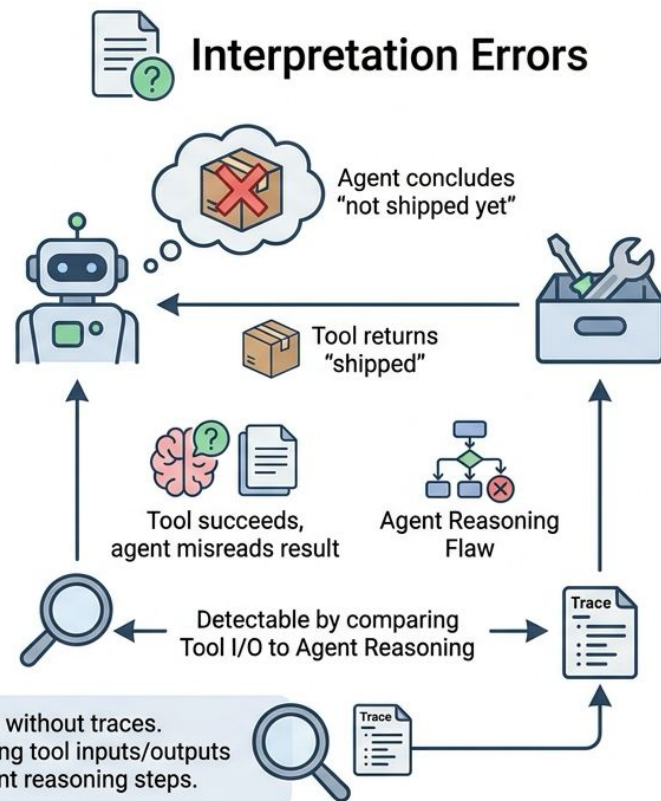
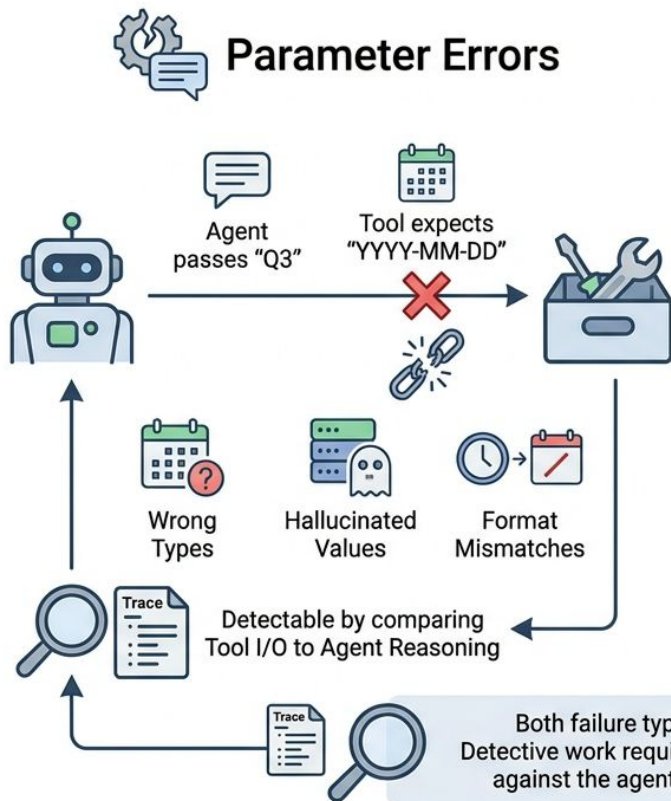
COMPARISON VIEWS



Forensic Debugging -- Tool Selection Failures

- Agent picks the wrong tool despite correct reasoning
 - Vague tool descriptions cause ambiguous selection
 - Trace reveals the thought, the action, and the mismatch
 - Fix: explicit use-case documentation per tool
 - Example: "search_sec_filings" vs. "search_general_web"
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- A decorative graphic in the bottom right corner consisting of several overlapping, semi-transparent blue rectangles and triangles, creating a modern, abstract look.

Debugging -- Parameters and Interpretation

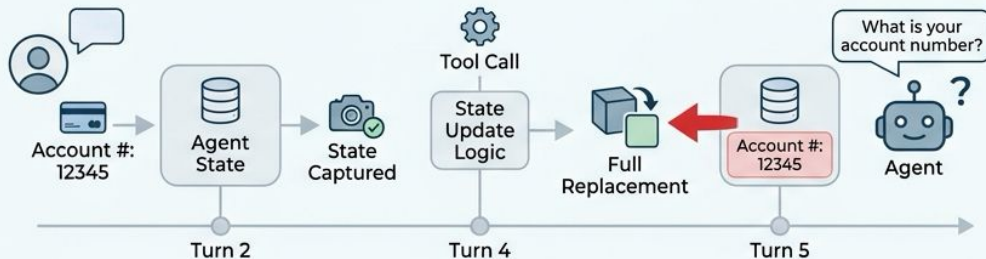


Both failure types are silent without traces.
Detective work requires comparing tool inputs/outputs
against the agent's subsequent reasoning steps.

State Management and Hallucination Detection

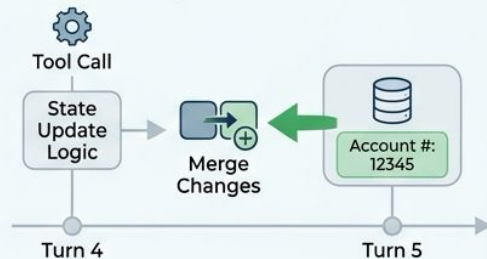
STATE MANAGEMENT FAILURE (AMNESIA) & FIX

Problem: State Loss via Overwrite



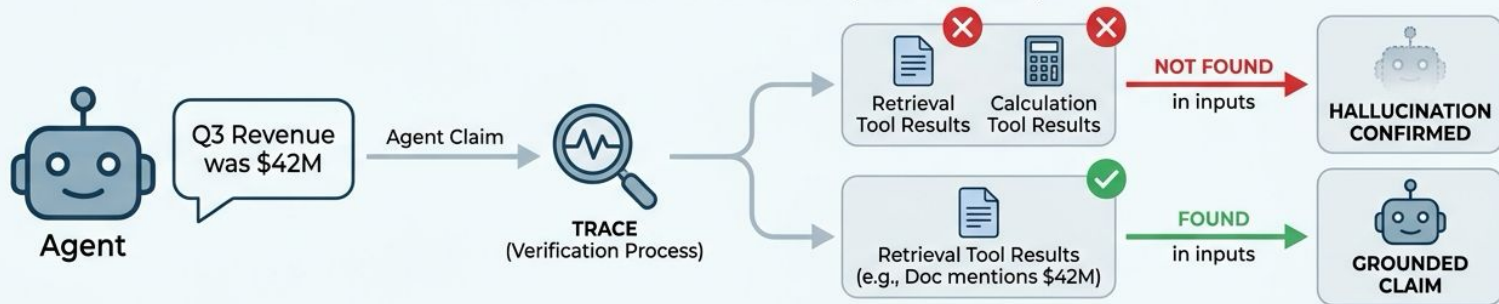
Outcome: Agent Forgets Information

Fix: Merge-Based Updates



Outcome: Information Preserved

HALLUCINATION DETECTION (FABRICATION)



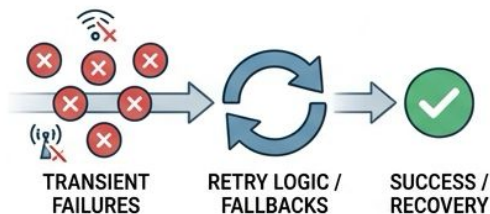
Correlate agent output against tool results in the trace to distinguish grounded claims from fabricated ones.

Production Observability Infrastructure

- OpenTelemetry standards integrate with enterprise platforms
 - Compatible backends: Jaeger, Prometheus, Langfuse, Phoenix
 - Architecture: agent SDK to Collector to backend storage
 - Centralized traces enable cross-agent correlation
 - Historical analysis detects regressions from model updates
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Distinguishing Signal from Noise

EXPECTED VS. UNEXPECTED ERRORS

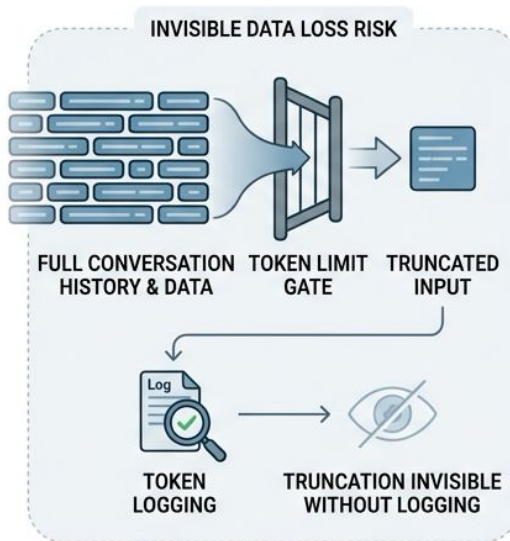


EXPECTED ERRORS: HANDLED



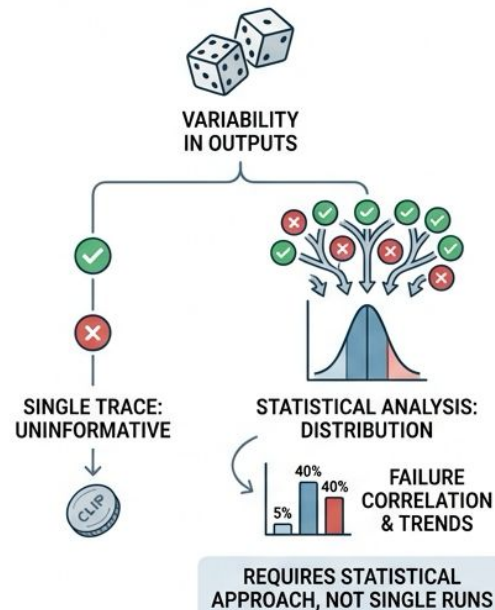
UNEXPECTED ERRORS: UNHANDLED

CONTEXT WINDOW TRUNCATION



INVISIBLE DATA LOSS RISK

NON-DETERMINISM & STATISTICAL ANALYSIS



Building a Trace-First Culture

- Instrument from day one, not after production incidents
 - Five-step debugging workflow: locate, examine, drill, inspect, trace
 - Weekly trace review sessions build team debugging skills
 - Goal: predictive debugging catching problems before users notice
 - Next chapter: tool call auditing validates what agents actually do
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